Q (i) $P = (\{q_0, q_1\}, \{0,1\}, \{0,1,Z_0\}, \delta, q_0, Z_0, \{q_1\})$

where δ is defined as,

$\delta(q_0, 0, Z_0) = \{(q_0, 0Z_0)\}$

$\delta(q_0, 1, Z_0) = \{(q_0, 1Z_0)\}$

$\delta(q_0, 0, 0Z_0) = \{(q_0, 00)\}$

$\delta(q_0, 1, 0Z_0) = \{(q_0, 10)\}$

$\delta(q_0, 0, 1Z_0) = \{(q_0, 01)\}$

$\delta(q_0, 1, 1Z_0) = \{(q_0, 11)\}$

$\delta(q_0, 0, 00) = \{(q_0, 000)\}$

$\delta(q_0, 0, 01) = \{(q_0, \epsilon)\}$

$\delta(q_0, a, 10) = \{(q_0, \epsilon)\}$

$\delta(q_0, 0, 11) = \{(q_0, 011)\}$

$\delta(q_0, 1, 00) = \{(q_0, \epsilon)\}$

$\delta(q_0, 1, 01) = \{(q_0, 101)\}$

$\delta(q_0, 1, 10) = \{(q_0, 110)\}$

$\delta(q_0, 1, 11) = \{(q_0, 111)\}$

$\delta(q_0, \epsilon, Z_0) = \{(q_1, Z_0)\}$

(c) The hating problem can be defined as $H(M)$ for a turing machine M such that M halts for a given input w (regardless of whether M accepts it or not). Then we need to find such pairs (M, w) such that w is in $H(M)$

Halting problem recursively enumerable:

We construct a universal touring machine - M_H such that it takes as input a pair (M, w) where M is a turing machine and w is a ~~bi~~ input for M.

(i) then if M hats then M_H accepts the input (M, w) and hafts (whether M acepts or rejects)

(ii) Otherwise if M doesn't halt, M_H doesn't halt too

Since this turing machine can be built; then halting problem is RE

Halting problem not recursive:

Let us say M is recursive.

Then We can prove this by building another touring machine M_2 that receives (M,w) as input.

we disign M_2 in such a way that if

(i) M halts, M_2 doesn't halt

(ii) M doesn't halt, M_2 halts

Now,

(i) if M halts the M_2 can exectute this input and won't halt. so, M won't halt, which is a contradiction

(ii) If M doesn't halt, then M_2 cannot execute inpu[t]

so, M_2 won't halt but If M doesn't halt then M_2 must halt which is also a contradiction.

so, assumption was wrong.

So, M is not recursive.